

Small-prime congruences for OEIS A348410

The cubic prime-power tower, an exact binary
defect formula, and a sharp denominator-12 theorem

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Abstract

Let $a(n) = [x^n]((1-x)(1-x^2))^{-n}$, the basket sequence OEIS A348410, and more generally $A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}$ for arbitrary integer parameters. An OEIS comment of Peter Bala conjectures the cubic congruence $a(mp^r) \equiv a(mp^{r-1}) \pmod{p^{3r}}$ for primes $p \geq 5$.

We prove two theorems. First, for every odd prime p and every N divisible by p , the difference $A_{\alpha,\beta}(N) - A_{\alpha,\beta}(N/p)$ is divisible by $p^{3v_p(N) - \varepsilon_p}$, where $\varepsilon_3 = 1$ and $\varepsilon_p = 0$ for $p \geq 5$; no coprimality restriction on the multiplier is needed. This contains Bala's assertion. A prior public proof candidate for the same odd-prime statement was located during the source audit and is credited below; no priority is claimed for it. Second, at the prime left untreated in that source we prove an explicit first binary defect congruence modulo 2^{3r-2} , which gives a sharp universal modulus 2^{3r-3} , improved to 2^{3r-2} when $\alpha\beta$ is even.

Together these yield $12n^{-3} \sum_{d|n} \mu(d)a(n/d) \in \mathbb{Z}_{\geq 0}$ for every $n \geq 1$, with 12 the smallest possible common denominator, and a primitive-necklace divisibility theorem. We isolate the exact hypothesis the logarithmic method needs, exhibit an infinite period-three family for which it fails modulo 5^3 , audit a printed framing statement that would have implied the contrary, derive algebraic generating functions and two correction terms in the asymptotic expansion, and supply exact reproducible checks. Mathematical completeness of the written proofs and global priority are separate matters: no claim of exhaustive novelty verification or external peer review is made.

Provenance. This article merges two independently produced research reports on OEIS A348410: **a348410-small-prime-congruences**, whose subject was the binary completion, the denominator theorem, the Möbius and asymptotic material, and the source audit; and **a348410-cubic-supercongruence**, whose subject was the odd-prime prime-power tower, its consequences for related families, and an audit of a broader printed framing claim. The shared odd-prime theorem (Theorem 1.2) was proved by the same argument in both reports and is therefore proved once here. Where the two reports proved the same statement by genuinely different routes — the $p = 3$ case of Lemma 3.3, the period-three obstruction of Section 7, and the framing audit of Appendix C — both arguments are preserved and marked as such.

Archive contents. This article in L^AT_EX and PDF; three verification programs; exact CSV tables; test reports; build instructions; a source and priority audit; BibTeX metadata; and a draft OEIS comment with a proof outline. No computational conjecture is used as a premise of a theorem.

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1 The problem, the source boundary, and the results

1.1 The sequence, the conjecture, and the source boundary

The sequence A348410 counts nonnegative integer solutions of

$$n = \sum_{i=1}^n (u_i + 2v_i). \quad (1.1)$$

Equivalently, n indistinguishable objects are distributed among n labelled white baskets and n labelled black baskets, and the black occupancies are even. We set $a(0) = 1$. The OEIS entry records the coefficient formula

$$a(n) = [x^n] \frac{1}{(1-x)^n(1-x^2)^n} \quad (1.2)$$

and, in a comment dated 21 February 2022, Bala's conjecture

$$a(mp^r) - a(mp^{r-1}) \in p^{3r}\mathbb{Z} \quad (p \geq 5 \text{ prime}, m, r \geq 1). \quad (1.3)$$

The live entry still labels this a conjecture in the version inspected for this article [1].

That label is not a reliable certificate of present-day openness. In particular, the 2026 articles of Niu and Prodingar already discuss the algebraic generating function [5, 6]. More importantly for this investigation, a public repository note, *A cubic tower for a two-parameter coefficient family*, contains an elementary proof candidate for (1.3), for its full integer-parameter analogue, and for a 3^{3r-1} ternary bound [3]. Its parameter convention is $[x^n]((1+x)^u(1-x)^v)^n$; ours corresponds to $(u, v) = (-\beta, -\alpha)$. The full odd-prime parameter conjecture is also stated on A352373 [2]. The repository source explicitly leaves the binary case without an assertion.

Two questions are therefore treated here, and they have different status. The odd-prime statement is proved in full in Section 3, as a first-class theorem of this article with a complete elementary proof; it is *not* claimed as a new resolution, because the prior proof candidate exists. The binary statement is the following completion problem, which the checked sources leave open.

Problem 1.1 (Binary completion). For the integer-parameter family

$$A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}, \quad \alpha, \beta \in \mathbb{Z}, \quad (1.4)$$

does cubic growth of the divisibility exponent persist at $p = 2$, with a fixed loss independent of the level? Can one determine the first possible binary defect and obtain an optimal common denominator for the cubic Möbius transform?

The sources checked did not supply the binary formula or the denominator conclusions below. This is a statement about the checked source boundary, not an exhaustive claim about all mathematical literature. Priority for the odd-prime result is credited to the prior proof candidate; the proof given here is independent of it, so that the binary completion does not rest on an external unreviewed argument.

1.2 Why the coefficient formula is the right handle

The displayed coefficient formula (1.2) is far more useful for divisibility questions than a high-order recurrence in n . Its logarithmic derivative has a coefficient sequence of period two, and multiplication of an index by an odd prime preserves parity. That single observation

converts the problem into an *exact* factorization, (3.8) below, rather than a congruence whose error has to be estimated. After the factorization, the only nontrivial remaining issue is whether a weighted reciprocal-square sum supplies one more full prime-power factor. Sections 3 and 7 show that it does in this setting, and that the corresponding statement for period three is false.

At $p = 2$ the parity argument collapses, because multiplication by 2 does not preserve parity. Section 4 replaces it by a reflection identity and a formal integration by parts.

1.3 Main theorems

All binomial coefficients with an integer upper argument use the generalized convention

$$\binom{z}{k} = \frac{z(z-1)\cdots(z-k+1)}{k!}, \quad k \geq 0. \quad (1.5)$$

In particular, this is an integer when z is a negative integer. For an odd prime p put

$$\varepsilon_p = \begin{cases} 1, & p = 3, \\ 0, & p \geq 5. \end{cases} \quad (1.6)$$

Theorem 1.2 (Odd-prime coefficient tower). *Let $\alpha, \beta \in \mathbb{Z}$, let p be an odd prime, and let $N > 0$ be divisible by p . Write $r = v_p(N)$. Then*

$$A_{\alpha,\beta}(N) - A_{\alpha,\beta}(N/p) \in p^{3r-\varepsilon_p}\mathbb{Z}. \quad (1.7)$$

Theorem 1.2 is proved in Section 3. Because the exponent is governed by the exact valuation of N rather than by a declared level, it contains the usual formulation with no coprimality restriction on the multiplier.

Corollary 1.3 (Prime-power form without coprimality). *For all $m, k \geq 1$ and every odd prime p ,*

$$A_{\alpha,\beta}(mp^k) \equiv A_{\alpha,\beta}(mp^{k-1}) \pmod{p^{3(k+v_p(m))-\varepsilon_p}}. \quad (1.8)$$

In particular $a(mp^r) \equiv a(mp^{r-1}) \pmod{p^{3r}}$ for $p \geq 5$, which is Bala's assertion (1.3), and $a(m3^r) \equiv a(m3^{r-1}) \pmod{3^{3r-1}}$.

Proof. Apply Theorem 1.2 to $N = mp^k$, whose exact valuation is $k + v_p(m)$. □

Theorem 1.4 (First binary defect). *Let $\alpha, \beta \in \mathbb{Z}$, let $m \geq 1$ be odd, and let $r \geq 2$. Then*

$$\begin{aligned} & A_{\alpha,\beta}(m2^r) - A_{\alpha,\beta}(m2^{r-1}) \\ & \equiv 2^{3r-3} \alpha \beta \binom{(\alpha + \beta + 1)m - 1}{m - 1} \pmod{2^{3r-2}}. \end{aligned} \quad (1.9)$$

Consequently the difference is always divisible by 2^{3r-3} , and is divisible by 2^{3r-2} whenever $\alpha\beta$ is even. If both parameters are odd and $m = 1$, then its valuation is exactly $3r - 3$, at every $r \geq 2$.

The first lift $r = 1$ is deliberately not included in the displayed defect formula. It always satisfies the elementary congruence modulo 2 proved below. The distinction prevents a common error: extrapolating a later-level binary estimate to the first lift.

Theorem 1.5 (All primes for the basket sequence). *For every prime p and all $m, r \geq 1$,*

$$v_p(a(mp^r) - a(mp^{r-1})) \geq 3r - v_p(12). \quad (1.10)$$

Thus the moduli are p^{3r} for $p \geq 5$, 3^{3r-1} for $p = 3$, and 2^{3r-2} for $p = 2$. Each of these uniform losses is necessary.

For a zero difference, its valuation is interpreted as $+\infty$.

Theorem 1.6 (Sharp common denominators). *Define*

$$B_{\alpha,\beta}(n) = \frac{1}{n^3} \sum_{d|n} \mu(d) A_{\alpha,\beta}(n/d), \quad B(n) = B_{2,1}(n). \quad (1.11)$$

Then $24B_{\alpha,\beta}(n) \in \mathbb{Z}$ for all integer parameters and all $n \geq 1$. If $\alpha\beta$ is even, $12B_{\alpha,\beta}(n) \in \mathbb{Z}$. In particular,

$$\boxed{\frac{12}{n^3} \sum_{d|n} \mu(d) a(n/d) \in \mathbb{Z}_{\geq 0} \quad (n \geq 1).} \quad (1.12)$$

The constant 12 is minimal for the single sequence $a(n)$. The constant 24 is minimal for the unrestricted integer-parameter family.

For the combinatorial interpretation, let P_n be the number of primitive cyclic words of length n in the weighted alphabet of pairs (u, v) of nonnegative integers, where (u, v) has weight $u + 2v$, and the total word weight is n . “Primitive” means that no shorter word is repeated to form it. Cyclic rotations are identified.

Corollary 1.7 (Primitive-orbit divisibility). *For every $n \geq 1$,*

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) a(n/d), \quad \frac{n^2}{\gcd(n^2, 12)} \mid P_n. \quad (1.13)$$

1.4 What is proved, and what is not claimed

All four displayed results — Theorems 1.2, 1.4, 1.5 and 1.6, together with Corollaries 1.3 and 1.7 — have complete elementary proofs here, in Sections 3, 4 and 5. They are theorems of this article, not statements imported from elsewhere.

Novelty is a separate question, and the answer differs by result. For the odd-prime tower, Theorem 1.2, a prior public elementary proof candidate for the same statement was located during the source audit [3]; no priority is claimed, and the proof here is given because the binary completion depends on it and should not rest on an external unreviewed argument. For the binary defect formula and the sharp denominators, nothing equivalent was found in the checked sources; that is a bounded observation about the sources checked, not a global priority claim. The algebraic generating function and the leading asymptotic are not claimed as new: both are already documented in the OEIS and related literature [1, 5, 6]. The correction coefficients and the general Gaussian-moment expression are derived explicitly here, without a priority claim. That a general printed framing statement has a problem is likewise not claimed as new; a period-four counterexample is already documented elsewhere [4], and the two period-three witnesses of Section 7 and Appendix C are compact independent boundary evidence. A stronger exact valuation pattern at powers of two is labelled as conjectural in Section 10.

The proofs have been checked algebraically and by independent exact computations, but have not been externally refereed or formalized in Lean. The code is a reproducibility aid, not a substitute for the arguments. No OEIS edit, database submission, or communication with an author has been performed.

2 Counting formulas and exact coefficient computation

2.1 A positive binomial sum

A single white basket contributes $(1 - x)^{-1}$ and a black basket contributes $(1 - x^2)^{-1}$. Multiplying over the labelled baskets proves (1.2). Expanding the two factors separately gives, for $n \geq 1$,

$$a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{2n-2k-1}{n-2k} \binom{n+k-1}{k}. \quad (2.1)$$

The first terms are

$$1, 1, 5, 19, 85, 376, 1715, 7890, 36693, 171820, 809380.$$

Every summand in (2.1) is nonnegative. For instance at $n = 5$ the three summands are 126, 175 and 75, giving $a(5) = 376$. This positive sum gives an independent check on the faster coefficient recurrence below. It also makes integrality and nonnegativity immediate for the target sequence.

For arbitrary integer parameters, generalized binomial expansion shows that

$$A_{\alpha,\beta}(n) = \sum_{k=0}^n (-1)^k \binom{\beta n + k - 1}{k} \binom{\alpha n + n - k - 1}{n - k} \quad (n \geq 1). \quad (2.2)$$

This formula includes zero and negative parameters without any change of convention. We put $A_{\alpha,\beta}(0) = 1$.

2.2 A two-step recurrence in the coefficient index

Fix n and write

$$F(x) = (1 - x)^{-\alpha n} (1 + x)^{-\beta n} = \sum_{k \geq 0} f_k x^k.$$

Logarithmic differentiation gives

$$(1 - x^2)F'(x) = n((\alpha - \beta) + (\alpha + \beta)x)F(x).$$

Consequently, with $f_{-1} = 0$ and $f_0 = 1$,

$$k f_k = n(\alpha - \beta) f_{k-1} + (n(\alpha + \beta) + k - 2) f_{k-2}, \quad k \geq 1. \quad (2.3)$$

The desired output is f_n . This is a recurrence in k with n fixed, not a recurrence relating successive diagonal values $A_{\alpha,\beta}(n)$.

The algorithm uses $O(n)$ exact arithmetic operations and two live integer coefficients. The integers grow with n , so this is not a claim of linear bit complexity. Every division by k is checked for exactness in the supplied implementation. Large datasets are computed without floating-point arithmetic.

One caveat matters for anyone reusing (2.3) in a divisibility experiment. The division by k is an exact division of integers, and it must not be replaced by multiplication by a modular inverse of k when $p \mid k$: no such inverse exists, and silently working modulo a prime power destroys exactly the p -adic information the congruences are about.

2.3 Small-prime boundaries visible in the first terms

For the target sequence,

$$\begin{aligned} a(2) - a(1) &= 4, \\ a(3) - a(1) &= 18 = 2 \cdot 3^2, \\ a(4) - a(2) &= 80 = 5 \cdot 2^4, \\ a(5) - a(1) &= 375 = 3 \cdot 5^3, \\ a(7) - a(1) &= 7889 = 23 \cdot 7^3. \end{aligned}$$

These examples already show that $p = 2$ and $p = 3$ cannot simply be inserted into Bala's stated modulus. They also prove the necessity of the fixed losses in Theorem 1.5: at $p = 2$ use level $r = 2$, and at $p = 3$ use level $r = 1$. At $p = 5$ there is no extra uniform power available. These are sharpness statements about the *uniform* formulas; they do not assert equality of the valuation at every prime, multiplier and level.

Exact valuations along the first few odd prime-power towers are collected in Table 1. Every entry is computed from integers, not inferred from a decimal approximation. The dash means

Table 1: Observed valuations $v_p(a(p^r) - a(p^{r-1}))$ at $m = 1$. The proved lower bound is $3r - \varepsilon_p$.

p	$r = 1$	$r = 2$	$r = 3$	$r = 4$
3	2	5	8	11
5	3	6	9	12
7	3	6	9	12
11	3	6	9	–

that the index 11^4 lies outside the default computational sweep, not that the congruence is unknown there. As a larger explicit odd check,

$$a(25) - a(5) = 13\,859\,743\,127\,937\,500$$

has 5-adic valuation 6, in agreement with the bound at $r = 2$. The analogous binary table is deferred to Section 10, because there the observed pattern is finer than anything proved here.

3 Local arithmetic and the odd-prime theorem

This section proves Theorem 1.2 in full. The argument is elementary and self-contained: formal logarithms over $\mathbb{Q}$, an exact factorization, one truncation of a formal exponential, and two coefficient estimates. It uses neither the algebraic generating function of Section 6 nor any general framing theorem; in particular it does not depend on the printed rational-framing statement discussed in Section 7 and Appendix C, whose explicit conclusion is false. The same statement has a prior public proof candidate [3], which is credited and not superseded; the proof below is independent of it.

3.1 The local ring and a coefficient estimate

For a prime p , let

$$R_p = \mathbb{Z}_{(p)} = \{u/v \in \mathbb{Q} : p \nmid v\}.$$

Congruence modulo p^h in this ring means that the difference has valuation at least h . A formal power-series congruence is always coefficientwise. All exponentials and logarithms are formal over $\mathbb{Q}$; at a fixed degree only finitely many terms contribute.

Set

$$\Phi(x) = (1-x)^{-\alpha}(1+x)^{-\beta}, \quad L(x) = \log \Phi(x) = \sum_{j \geq 1} \frac{c_j}{j} x^j, \quad c_j = \alpha + \beta(-1)^j. \quad (3.1)$$

Lemma 3.1 (Logarithmic-derivative estimate). *Let $G \in 1 + xR_p[[x]]$ and write*

$$x \frac{G'(x)}{G(x)} = \sum_{i \geq 1} c_i x^i, \quad c_i \in R_p.$$

Let $M \geq 1$ and put $g_j = [x^j]G(x)^M$. Then $g_0 = 1$, every $g_j \in R_p$, and for $j \geq 1$,

$$v_p(g_j) \geq \max\{0, v_p(M) - v_p(j)\}. \quad (3.2)$$

If moreover $G = \Phi$ as in (3.1) with integer parameters, then $g_j \in \mathbb{Z}$.

Proof. Differentiating G^M and multiplying by x gives $x(G^M)' = M(xG'/G)G^M$, that is, the exact identity

$$jg_j = M \sum_{i=1}^j c_i g_{j-i}. \quad (3.3)$$

The sum on the right lies in R_p . Thus $v_p(g_j) \geq v_p(M) - v_p(j)$. Independently $G^M \in R_p[[x]]$, which gives the lower bound zero and hence the maximum. Integrality of g_j in the case $G = \Phi$ follows from generalized binomial expansion. $\square$

The estimate is useful precisely when a complementary index is more divisible by p than j itself, which is how it will be applied. No asymptotics and no induction on a guessed recurrence are involved. Stating the lemma for a general G costs nothing and is what Remark 3.7 below needs.

The second input is a truncation of the formal exponential.

Lemma 3.2 (Exponential truncation). *Let p be an odd prime, let N be a positive integer with $r = v_p(N) \geq 1$, and let $W \in xR_p[[x]]$. Then*

$$\exp(NW) \equiv 1 + NW + \frac{N^2}{2}W^2 \pmod{p^{3r-\varepsilon_p}R_p[[x]]}. \quad (3.4)$$

Proof. Each coefficient of $(N^k/k!)W^k$ has valuation at least $kr - v_p(k!)$. Writing k in base p gives the elementary estimate

$$v_p(k!) = \sum_{i \geq 1} \left\lfloor \frac{k}{p^i} \right\rfloor \leq \frac{k-1}{p-1}.$$

For $p \geq 5$ this yields $v_p(k!) \leq (k-1)/4 \leq k-3$ when $k \geq 4$, and the same bound holds directly at $k = 3$ because $v_p(3!) = 0$. Hence for $k \geq 3$,

$$kr - v_p(k!) \geq 3r + (k-3)(r-1) \geq 3r.$$

For $p = 3$ the estimate reads $v_3(k!) \leq (k-1)/2 \leq k-2$, so

$$kr - v_3(k!) \geq 3r - 1 + (k-3)(r-1) \geq 3r - 1.$$

In both cases every omitted term lies in the stated ideal, coefficient by coefficient. $\square$

3.2 A reciprocal-square cancellation

Recall the constant ε_p of (1.6): it is 1 at $p = 3$ and 0 for $p \geq 5$.

Lemma 3.3 (Reduced reciprocal squares). *If $s > 0$, $p \mid s$, and $t = v_p(s)$, then*

$$H_p(s) = \sum_{\substack{1 \leq j < s \\ p \nmid j}} \frac{1}{j^2} \in p^{t-\varepsilon_p} R_p. \quad (3.5)$$

Proof. Reduce denominators modulo p^t and partition the summation interval into complete blocks of length p^t . It is enough to consider $U_t = \sum_{u \in (\mathbb{Z}/p^t\mathbb{Z})^\times} u^{-2}$.

For $p \geq 5$, multiplication by 2 permutes the units and gives $U_t \equiv 2^{-2}U_t \pmod{p^t}$. Hence $3U_t \equiv 0 \pmod{p^t}$; since 3 is invertible, $U_t \equiv 0$.

For $p = 3$, induction proves $U_t \equiv 0 \pmod{3^{t-1}}$. The case $t = 1$ has modulus 1. For $t \geq 2$, each unit modulo 3^{t-1} has three lifts, so $U_t \equiv 3U_{t-1} \pmod{3^{t-1}}$. The induction hypothesis finishes the proof. The same bounds hold for the full block sum. $\square$

Remark 3.4 (A second, uniform proof of Lemma 3.3). The case distinction can be removed entirely. Multiplication by 2 permutes $(\mathbb{Z}/p^t\mathbb{Z})^\times$ for *every* odd p , so

$$U_t \equiv \sum_{a \in (\mathbb{Z}/p^t\mathbb{Z})^\times} (2a)^{-2} \equiv \frac{1}{4}U_t \pmod{p^t},$$

whence $3U_t \equiv 0 \pmod{p^t}$ and therefore $v_p(U_t) \geq t - v_p(3) = t - \varepsilon_p$, uniformly in the odd prime p . What the alternative route buys is that the exceptional prime is not an extra case at all: the loss of one power at $p = 3$ is visibly the valuation of the single constant 3 produced by the doubling automorphism, rather than the outcome of a separate lifting induction. It also shows why the *unweighted* sum is safe to permute in the first place: its summand depends only on the unit residue modulo p^t . The first proof is retained because its lifting induction is the argument that generalizes when the doubling automorphism is unavailable, as at $p = 2$ in Lemma 4.1.

For odd p , parity is preserved by multiplication by p , so $c_{pj} = c_j$. The reduced logarithm

$$D_p(x) = L(x) - \frac{1}{p}L(x^p) = \sum_{\substack{j \geq 1 \\ p \nmid j}} \frac{c_j}{j} x^j \quad (3.6)$$

belongs to $xR_p[[x]]$ and has no monomial whose exponent is divisible by p .

Lemma 3.5 (Quadratic-tail estimate). *If $s > 0$ is divisible by the odd prime p , then*

$$T_p(s) := [x^s]D_p(x)^2 \in p^{v_p(s)-\varepsilon_p} R_p. \quad (3.7)$$

Proof. Write $t = v_p(s)$. Since $p \mid s$, both j and $s - j$ are units whenever $p \nmid j$. Thus

$$T_p(s) = \sum_{\substack{1 \leq j < s \\ p \nmid j}} \frac{c_j c_{s-j}}{j(s-j)} \equiv - \sum_{\substack{1 \leq j < s \\ p \nmid j}} \frac{c_j c_{s-j}}{j^2} \pmod{p^t}.$$

If s is odd, the weight $c_j c_{s-j} = \alpha^2 - \beta^2$ is constant, so Lemma 3.3 applies directly.

If s is even, the weights are $(\alpha + \beta)^2$ for even j and $(\alpha - \beta)^2$ for odd j . The weighted reciprocal-square sum is exactly

$$(\alpha - \beta)^2 H_p(s) + \alpha\beta H_p(s/2).$$

Indeed, the excess even weight is $4\alpha\beta$, while replacing j by $2k$ divides the reciprocal square by 4. Since p is odd, $v_p(s/2) = t$. Applying Lemma 3.3 to both terms proves the claim. $\square$

Remark 3.6 (Why the parity split, and not a permutation). The parity decomposition in the even case is the essential structural step, and it cannot be replaced by an appeal to periodicity. Lemma 3.3 is proved by permuting the summands of an *unweighted* sum, which is legitimate because its summand depends only on the residue of j modulo p^t . A weighted sum admits no such permutation: reducing an index modulo p^t can change its residue modulo the period of the weights, so a rearrangement that fixes the reciprocal squares need not fix the weights. What makes the step valid here is the exact identity displayed above, which rewrites the weighted sum as a fixed linear combination of two unweighted ones, not a rearrangement of it. Section 7 exhibits a period-three weight for which no such identity is available and the conclusion genuinely fails.

3.3 Proof of the full odd-prime tower

Proof of Theorem 1.2. Put $N = pM$, so that $v_p(M) = r - 1$. From (3.6) one obtains, by splitting the logarithm at the degrees divisible by p , the exact identity

$$\Phi(x)^N = \Phi(x^p)^M \exp(ND_p(x)). \quad (3.8)$$

This is an identity in $\mathbb{Q}[[x]]$, not a congruence whose error has to be estimated. Lemma 3.2, applied with $W = D_p$, gives

$$\exp(ND_p) \equiv 1 + ND_p + \frac{N^2}{2}D_p^2 \pmod{p^{3r-\varepsilon_p}R_p[[x]]}, \quad (3.9)$$

and thereby justifies every discarded term of the formal exponential.

Extract degree N . The constant term of the exponential gives $A_{\alpha,\beta}(M)$. The linear contribution is zero: $\Phi(x^p)^M$ is supported on exponents divisible by p , whereas D_p is supported away from them. Therefore

$$A_{\alpha,\beta}(N) - A_{\alpha,\beta}(M) \equiv \frac{N^2}{2}[x^N]\Phi(x^p)^M D_p(x)^2 \pmod{p^{3r-\varepsilon_p}}. \quad (3.10)$$

Writing $g_j = [x^j]\Phi(x)^M$, the remaining coefficient is

$$\sum_{j=0}^{M-1} g_j T_p(p(M-j)). \quad (3.11)$$

We claim that each summand belongs to $p^{r-\varepsilon_p}R_p$. For $j = 0$, this follows directly from Lemma 3.5 at $s = N$. For $j > 0$ with $v_p(j) < r - 1$, the equality $v_p(M) = r - 1$ gives $v_p(M - j) = v_p(j)$. Lemmas 3.1 and 3.5 then supply, respectively,

$$r - 1 - v_p(j) \quad \text{and} \quad 1 + v_p(j) - \varepsilon_p,$$

whose sum is $r - \varepsilon_p$. For $v_p(j) \geq r - 1$, we have $v_p(M - j) \geq r - 1$, and the quadratic-tail bound alone supplies $r - \varepsilon_p$.

Finally, $v_p(N^2/2) = 2r$ because p is odd. Inserting the valuation of (3.11) into (3.10) proves (1.7). The difference is an ordinary integer, so its local divisibility is ordinary integer divisibility. $\square$

With Theorem 1.2 proved, Corollary 1.3 follows as stated in Section 1: substituting $N = mp^k$, whose exact valuation is $k + v_p(m)$, gives (1.8) with no coprimality restriction on m .

3.4 Where the three powers come from

The exponent $3r$ is not a numerical coincidence, and it is worth recording how it is assembled. Two of the three powers of p come from the scalar $N^2/2$ in front of the quadratic exponential term. The third is produced jointly by the two factors of a single summand of (3.11). For the summand indexed by j with $t = v_p(j) < r - 1$, the quadratic-tail estimate of Lemma 3.5 at $s = p(M - j)$ contributes $1 + t - \varepsilon_p$ powers, which is less than $r - \varepsilon_p$; exactly the shortfall $r - 1 - t$ is then supplied by the coefficient g_j through Lemma 3.1. The two estimates are complementary: whichever of the two indices j and $M - j$ carries little p -divisibility, the other carries the rest.

This is also the reason a proof written only for $N = p$ would miss an essential part of the argument. At $N = p$ one has $M = 1$, the sum (3.11) has the single term $j = 0$, and the coefficient estimate of Lemma 3.1 is never used. The interaction between the two estimates, which is what makes the tower climb, appears only at level two and beyond.

Remark 3.7 (A sufficient criterion beyond this family). Fix a prime $p \geq 5$. The proof above used two properties of Φ and nothing else. Accordingly, let $G \in 1 + xR_p[[x]]$ be any series whose logarithmic-derivative coefficients $xG'/G = \sum_{j \geq 1} c_j x^j$ satisfy $c_{pj} = c_j$, and put $A(n) = [x^n]G(x)^n$. If in addition

$$[x^s] \left(\sum_{p \nmid j} \frac{c_j}{j} x^j \right)^2 \in p^{v_p(s)} R_p \quad \text{for every } s \text{ with } p \mid s, \quad (3.12)$$

then $A(N) - A(N/p) \in p^{3v_p(N)} R_p$ for every N divisible by p . The proof is verbatim that of Theorem 1.2, with Lemma 3.5 replaced by the hypothesis (3.12) and Lemma 3.1 applied to G in its general form.

The point of isolating the criterion is the word “in addition”. Exact invariance $c_{pj} = c_j$ under index multiplication gives the factorization and kills the linear term, but it does not give (3.12): the quadratic estimate is a genuinely additional hypothesis, not a consequence of periodicity or of Frobenius invariance. Section 7 exhibits a period-three sequence satisfying $c_{5j} = c_j$ for which (3.12) fails at $p = 5$, and with it the conclusion.

3.5 Other integer-sequence families

Theorem 1.2 covers more than the basket sequence, and a few reparametrizations make the range visible.

Since $(1 - x)(1 + x) = 1 - x^2$, a change of parameters turns the family into a two-kind basket family. For integers σ, τ ,

$$A_{\sigma+\tau, \tau}(n) = [x^n](1 - x)^{-\sigma n}(1 - x^2)^{-\tau n}. \quad (3.13)$$

For nonnegative σ, τ the right side counts distributions of n objects into σn labelled unrestricted baskets and τn labelled even-occupancy baskets; the case $\sigma = \tau = 1$ is $A_{2,1} = a$, that is, (1.1). All of these sequences obey the same odd-prime congruences. Negative σ or τ remains covered algebraically through (2.2), but the basket reading is then unavailable.

Two elementary specializations are useful as sanity checks. For a positive integer α ,

$$A_{\alpha,0}(n) = \binom{(\alpha+1)n-1}{n} \quad (n \geq 1), \quad (3.14)$$

by the generalized hockey-stick identity. For a positive integer q , $\Phi = (1 - x^2)^{-q}$ when $\alpha = \beta = q$, so

$$A_{q,q}(n) = \begin{cases} 0, & n \text{ odd}, \\ \binom{qn + n/2 - 1}{n/2}, & n \text{ even}. \end{cases} \quad (3.15)$$

Theorem 1.2 accommodates the vanishing odd terms and the nonzero even ones without a separate divisibility convention, since 0 is divisible by every power of p .

Finally, the step of the second factor may be changed freely away from its own prime divisors.

Corollary 3.8 (Change of step). *Let $d \geq 1$ and $\alpha, \beta \in \mathbb{Z}$, and put*

$$A^{(d)}(n) = [x^n](1 - x^d)^{-\alpha n}(1 + x^d)^{-\beta n}.$$

Let p be an odd prime with $p \nmid d$, and let $N > 0$ be divisible by p . Then

$$A^{(d)}(N) - A^{(d)}(N/p) \in p^{3v_p(N) - \varepsilon_p} \mathbb{Z}.$$

Proof. The series $(1 - x^d)^{-\alpha n}(1 + x^d)^{-\beta n}$ is supported on exponents divisible by d . Since $p \nmid d$, the conditions $d \mid N$ and $d \mid N/p$ are equivalent. If they fail, both coefficients vanish and the difference is 0. If they hold, then $A^{(d)}(N) = A_{\alpha,\beta}(N/d)$ and $A^{(d)}(N/p) = A_{\alpha,\beta}(N/(pd))$, while $v_p(N/d) = v_p(N)$ because $p \nmid d$. Theorem 1.2 applied at N/d gives the claim. $\square$

The hypothesis $p \nmid d$ is not decorative, and the restriction to the divisors of d is not the only one that matters: Section 7 shows that replacing $1 + x^2$ by $1 - x^3$ — a change of *shape* rather than of *step* — destroys the conclusion at $p = 5$, a prime that divides neither the old period nor the new one.

3.6 Limits along a prime-power tower

Fix an odd prime p , integer parameters, and a positive integer m . By Corollary 1.3 the differences $A_{\alpha,\beta}(mp^r) - A_{\alpha,\beta}(mp^{r-1})$ have valuations tending to infinity, so the sequence $r \mapsto A_{\alpha,\beta}(mp^r)$ is Cauchy in $\mathbb{Z}_p$ and converges to a limit $\lambda_{\alpha,\beta,p,m} \in \mathbb{Z}_p$. Summing the tail of successive differences gives the error estimate

$$v_p(\lambda_{\alpha,\beta,p,m} - A_{\alpha,\beta}(mp^r)) \geq 3(r + 1 + v_p(m)) - \varepsilon_p \quad (r \geq 0). \quad (3.16)$$

This is a consequence of the full tower, not of the first congruence alone, and it does not require identifying the limit in closed form. We make no claim here about what $\lambda_{\alpha,\beta,p,m}$ is.

4 The binary defect: proof and sharpness

At $p = 2$, the equality $c_{pj} = c_j$ used in (3.6) fails. The correct replacement is a reflection argument. It separates the even and odd parts before any exponential is truncated.

4.1 An exact odd-harmonic normalization

Work in $R_2 = \mathbb{Z}_{(2)}$ and define

$$U(x) = \sum_{\substack{j \geq 1 \\ j \text{ odd}}} \frac{x^j}{j} = \frac{1}{2} \log \frac{1+x}{1-x}. \quad (4.1)$$

The factor $1/2$ in the logarithmic expression is harmless: the displayed odd-power series has coefficients in R_2 .

Lemma 4.1 (Odd harmonic sums are exactly normalized). *For every positive even T , let*

$$h(T) = \sum_{\substack{1 \leq j < T \\ j \text{ odd}}} \frac{1}{j}.$$

Then

$$v_2(h(T)) = 2v_2(T) - 2. \quad (4.2)$$

In particular, $h(2j)/j^2$ is an odd element of R_2 for every $j \geq 1$.

Proof. Write $T = 2^e q$ with q odd. First,

$$\sum_{\substack{1 \leq u < 2^e \\ u \text{ odd}}} u^{-2} \equiv 2^{e-1} \pmod{2^e}. \quad (4.3)$$

For $e = 1$ this is immediate. For $e \geq 2$, the two lifts u and $u + 2^{e-1}$ of an odd residue modulo 2^{e-1} have equal inverse squares modulo 2^e . Indeed their squares differ by a multiple of 2^e . Thus the sum at level e is twice the sum at level $e - 1$ modulo 2^e , which proves (4.3) by induction.

Partitioning the interval $1 \leq j < T$ into q blocks now shows that $\sum_{j \text{ odd}} j^{-2} \equiv 2^{e-1} \pmod{2^e}$. Pair j with $T - j$ to obtain the exact identity

$$h(T) = \frac{T}{2} \sum_{\substack{1 \leq j < T \\ j \text{ odd}}} \frac{1}{j(T-j)}.$$

Modulo 2^e , the sum on the right is the negative of the reciprocal-square sum. It consequently has valuation exactly $e - 1$. The prefactor $T/2$ has valuation $e - 1$ as well, proving (4.2). $\square$

The even power series $U(x)^2$ satisfies

$$[x^{2j}]U(x)^2 = \sum_{\substack{1 \leq k < 2j \\ k \text{ odd}}} \frac{1}{k(2j-k)} = \frac{h(2j)}{j}.$$

Define

$$K(t) = \sum_{j \geq 1} \frac{h(2j)}{j^2} t^j \in tR_2[[t]], \quad V(t) = U(\sqrt{t})^2. \quad (4.4)$$

Here $U(\sqrt{t})^2$ denotes the integer-exponent series obtained from the even series $U(x)^2$ by replacing x^2 with t ; no choice of an analytic square root is involved. The name V is local to this section and is unrelated to the framing inputs V_1 and V_3 of Section 7 and Appendix C. The identities we need are

$$V(t) = tK'(t), \quad U(t)^2 = \frac{t}{2} \frac{d}{dt} K(t^2), \quad K(t) \equiv \frac{t}{1-t} \pmod{2}. \quad (4.5)$$

The first two follow by coefficient comparison. The last follows from the oddness assertion of Lemma 4.1. This final residue, not just a divisibility bound, will determine the leading binary digit.

4.2 Separating the two coefficient problems

Set

$$S = \alpha + \beta, \quad \delta = \alpha - \beta.$$

Let M be even, put $s = v_2(M) \geq 1$, and define

$$g(t) = (1 - t)^{-SM}.$$

The logarithmic identities

$$\begin{aligned} \Phi(x)^{2M} &= (1 - x^2)^{-SM} \exp(2\delta MU(x)), \\ \Phi(t)^M &= g(t) \exp(-2\beta MU(t)) \end{aligned}$$

are exact. Only the even part of the first exponential contributes to degree $2M$. Consequently

$$\begin{aligned} A_{\alpha,\beta}(2M) &= [t^M]g(t) \cosh(2\delta MU(\sqrt{t})), \\ A_{\alpha,\beta}(M) &= [t^M]g(t) \exp(-2\beta MU(t)). \end{aligned} \quad (4.6)$$

The hyperbolic cosine is interpreted as its even formal Taylor series, so every exponent of t in (4.6) is an integer.

Since $v_2(k!) \leq k - 1$, for $k \geq 3$ we have

$$v_2\left(\frac{(2M)^k}{k!}\right) \geq ks + 1 \geq 3s + 1.$$

The same estimate applies to all even degrees $k \geq 4$ in the cosine. Thus modulo 2^{3s+1} we need only three coefficient expressions:

$$I_1 = [t^M]gU, \quad I_2 = [t^M]gU^2, \quad J = [t^M]gV. \quad (4.7)$$

Writing $\Delta = A_{\alpha,\beta}(2M) - A_{\alpha,\beta}(M)$, expansion of (4.6) gives

$$\Delta \equiv 2\beta MI_1 + 2\delta^2 M^2 J - 2\beta^2 M^2 I_2 \pmod{2^{3s+1}}. \quad (4.8)$$

4.3 Reflection converts the linear term into a quadratic one

The apparent obstacle in (4.8) is the linear term I_1 . Because M is even and U is odd,

$$[t^M]g(-t)U(t) = -I_1.$$

Also

$$\frac{g(-t)}{g(t)} = \left(\frac{1-t}{1+t}\right)^{SM} = \exp(-2SMU(t)).$$

Combining these identities yields

$$I_1 = \frac{1}{2}[t^M]g(t)U(t) (1 - \exp(-2SMU(t))). \quad (4.9)$$

The first exponential term in (4.9) is SMI_2 . Every remaining term has a scalar coefficient with valuation at least $2s$: for $k \geq 2$,

$$v_2\left(\frac{(2SM)^k}{2k!}\right) \geq k(s+1) - 1 - v_2(k!) \geq ks \geq 2s.$$

Hence

$$I_1 \equiv SMI_2 \pmod{2^{2s}R_2}. \quad (4.10)$$

Multiplication by $2\beta M$ makes the error divisible by 2^{3s+1} . Substituting in (4.8) and using $\beta S - \beta^2 = \alpha\beta$ therefore gives

$$\Delta \equiv 2M^2(\alpha\beta I_2 + \delta^2 J) \pmod{2^{3s+1}}. \quad (4.11)$$

This reflection is the step that replaces the odd-prime support cancellation. It is valid because M is even; this is exactly why the first binary lift was separated from the theorem.

4.4 Formal integration by parts supplies the third power

For any $K_0(t) \in R_2[[t]]$, coefficient differentiation gives

$$[t^M]g(t)tK_0'(t) = M[t^M]g(t)K_0(t) \left(1 - \frac{St}{1-t}\right). \quad (4.12)$$

Indeed, $[t^M]t(gK_0)' = M[t^M]gK_0$ and $tg'/g = SMt/(1-t)$.

Using (4.5) in (4.12) shows

$$J \in MR_2, \quad I_2 = \frac{M}{2}C, \quad (4.13)$$

where

$$C = [t^M](1-t)^{-SM}K(t^2) \left(1 - \frac{St}{1-t}\right) \in R_2. \quad (4.14)$$

Thus the J term in (4.11) is already zero modulo 2^{3s+1} , and we obtain the particularly useful congruence

$$\Delta \equiv \alpha\beta M^3 C \pmod{2^{3s+1}}. \quad (4.15)$$

It immediately proves the universal valuation bound $3s$ and its one-bit improvement when $\alpha\beta$ is even.

It remains to compute C modulo 2 when both parameters are odd. In this case S is even. Write $M = 2^s m$ with m odd. Over $\mathbb{F}_2[[t]]$, generalized integer powers and the Frobenius identity give

$$(1-t)^{-SM} \equiv (1-t^{2^s})^{-Sm}, \quad 1 - \frac{St}{1-t} \equiv 1,$$

while $K(t^2) \equiv t^2/(1-t^2)$. Since $2^s \geq 2$, extraction of t^M then gives

$$\begin{aligned} C &\equiv \sum_{j=0}^{m-1} \binom{Sm+j-1}{j} \\ &= \binom{(S+1)m-1}{m-1} \pmod{2}. \end{aligned} \quad (4.16)$$

The last identity is the generalized hockey-stick identity, obtained by multiplying $(1-z)^{-Sm}$ by $(1-z)^{-1}$ and extracting degree $m-1$. It holds for every integer S , including negative values.

If $\alpha\beta$ is even, both sides of the desired leading-bit formula vanish regardless of C . If $\alpha\beta$ is odd, insert (4.16) into (4.15). Finally, $M^3 = 2^{3s}m^3 \equiv 2^{3s} \pmod{2^{3s+1}}$. Taking $s = r-1$ proves Theorem 1.4 in full. For $m = 1$ the binomial coefficient in (1.9) equals 1, so odd parameters give an odd leading quotient and exact valuation $3r-3$.

4.5 First lift, optimal losses, and the all-prime statement

For any $\Phi \in 1 + x\mathbb{Z}[[x]]$,

$$\Phi(x)^2 \equiv \Phi(x^2) \pmod{2}.$$

Raising this to m and extracting degree $2m$ proves

$$A_{\alpha,\beta}(2m) \equiv A_{\alpha,\beta}(m) \pmod{2}. \quad (4.17)$$

For $(\alpha, \beta) = (2, 1)$, the improved binary bound of Theorem 1.4 is $3r - 2$ for $r \geq 2$, and (4.17) supplies exactly the claimed lower bound 1 at $r = 1$. If m is even, apply the theorem at the larger exact binary level. Together with Theorem 1.2, this proves Theorem 1.5. The examples in Section 2 prove its uniform sharpness.

There is a distinction between uniform sharpness and eventual sharpness. For the target sequence, equality occurs at $r = 2$, but later powers of two show more divisibility. The theorem does not assert equality at every level. For the unrestricted family, in contrast, odd parameters and $m = 1$ attain the general bound $3r - 3$ at every level $r \geq 2$.

5 Möbius transforms, common denominators, and cyclic words

5.1 From a prime-power tower to a denominator theorem

For any sequence $A(n)$, set

$$T_A(n) = \sum_{d|n} \mu(d) A(n/d).$$

If $n = p^r n_0$ with $p \nmid n_0$, separating divisors according as they contain p gives the exact identity

$$T_A(n) = \sum_{d|n_0} \mu(d) \left(A(p^r n_0/d) - A(p^{r-1} n_0/d) \right). \quad (5.1)$$

Only squarefree divisors matter, so there are no terms containing a higher power of p in the divisor itself.

Apply (5.1) to the basket sequence. Theorem 1.5 implies, for every prime $p \mid n$,

$$v_p(T_a(n)) \geq 3v_p(n) - v_p(12).$$

Therefore $n^3 \mid 12T_a(n)$, proving the integrality part of (1.12).

For the general family, the odd-prime bounds are unchanged and the binary bound loses at most three powers. At level one the needed general estimate is merely valuation at least zero. Consequently

$$n^3 \mid 24T_{A_{\alpha,\beta}}(n).$$

If $\alpha\beta$ is even, the improved binary estimate and (4.17) reduce 24 to 12. This proves all the integrality assertions of Theorem 1.6.

The constants cannot be reduced. For the target sequence,

$$B(3) = \frac{19-1}{27} = \frac{2}{3}, \quad B(4) = \frac{85-5}{64} = \frac{5}{4}. \quad (5.2)$$

Every integer clearing all denominators must thus be divisible by both 3 and 4. Their least common multiple is 12. For the unrestricted family, $(\alpha, \beta) = (1, 1)$ gives

$$B_{1,1}(4) = \frac{10-2}{64} = \frac{1}{8},$$

and the target example at $n = 3$ forces a factor 3, so a universal clearing integer must be divisible by 24. Even the single parameter pair $(1, 3)$ requires 24: its values at $n = 3$ and $n = 4$ are $-8/3$ and $59/8$.

5.2 Primitive cyclic words and positivity

Let $\mathcal{A} = \{(u, v) : u, v \in \mathbb{Z}_{\geq 0}\}$, with weight $u + 2v$. A word of length n and total weight n corresponds exactly to a solution of (1.1). Although the alphabet is infinite, the weight constraint permits only finitely many letters at each fixed n .

A word of length n whose least period is $d \mid n$ is the repetition of a primitive block of length d . Its total weight is n precisely when the block weight is d . A primitive cyclic orbit of such blocks has exactly d distinct linear representatives. Therefore

$$a(n) = \sum_{d \mid n} dP_d. \quad (5.3)$$

Möbius inversion proves the first formula of (1.13) and also proves $P_n \geq 0$. Since $B(n) = P_n/n^2$, nonnegativity in Theorem 1.6 follows. Integrality of $12P_n/n^2$ is equivalent to the second assertion of (1.13).

The first values illustrate that this is stronger than the ordinary integrality of a necklace count:

n	P_n	$B(n) = P_n/n^2$	$12B(n)$
1	1	1	12
2	2	1/2	6
3	6	2/3	8
4	20	5/4	15
5	75	3	36
6	282	47/6	94
7	1127	23	276
8	4576	143/2	858
9	19089	707/3	2828
10	80900	809	9708
11	348238	2878	34536
12	1516620	126385/12	126385

For example, $P_7 = 49 \cdot 23$, and P_6 is divisible by $36/\gcd(36, 12) = 3$. The individual counts need not be multiples of n^2 ; small-prime denominators are genuine.

The entry at $n = 12$ deserves a remark. The minimality argument above uses two indices, $n = 3$ and $n = 4$, to force the factors 3 and 4 separately. The value $B(12) = 126385/12$ shows more: the full denominator 12 is already attained at a single index, so no proper divisor of 12 clears the transform even along one arithmetic progression.

5.3 Lambert series and an Euler product

Let $H(z) = \sum_{n \geq 0} a(n)z^n$. Equation (5.3) gives the formal Lambert expansion

$$H(z) - 1 = \sum_{n \geq 1} nP_n \frac{z^n}{1 - z^n} = \sum_{n \geq 1} n^3 B(n) \frac{z^n}{1 - z^n}. \quad (5.4)$$

All operations are valid formally because only finitely many divisors contribute to a fixed coefficient. The denominator theorem can be stated as follows: the cubic Lambert coefficients of $12(H - 1)$ are nonnegative integers, and 12 is the least positive integer with this property.

A related Euler product will follow from the Lagrange series in the next section:

$$\frac{y(z)}{z} = \prod_{n \geq 1} (1 - z^n)^{-P_n}. \quad (5.5)$$

To verify it directly, take formal logarithms. The coefficient of z^k on the right is $k^{-1} \sum_{n|k} n P_n = a(k)/k$, exactly the logarithm on the left derived below.

6 Algebraic generating functions

The main generating-function identities are established facts for A348410. Neither the parametric generating function, nor the quartic it satisfies, nor the extraction recurrence (2.3) is claimed as new: the first two are recorded in the OEIS entry [1] and treated by Niu [5] and Prodinger [6], and Prodinger also extracts a recurrence. We include short derivations because they connect the coefficient-power definition, the Euler product, and the asymptotic analysis, and because they make the symbolic checks transparent.

It is worth saying plainly what this section is *not* for. Algebraicity is not used to prove any supercongruence in this article. The logarithmic coefficient pattern of Φ , rather than the algebraicity of H , is the input that supplies the divisibility; Remark 3.7 and Section 7 show that the relevant hypothesis is a statement about $\log \Phi$, and that algebraic sequences of the same shape can fail it.

6.1 Lagrange reversion for the full parameter family

Let $y(z)$ be the unique series in $z\mathbb{Z}[[z]]$ satisfying

$$y = z\Phi(y). \quad (6.1)$$

Existence and uniqueness follow recursively from $\Phi(0) = 1$. Write $H_{\alpha,\beta}(z) = \sum_{n \geq 0} A_{\alpha,\beta}(n)z^n$. Lagrange inversion, applied to $\log \Phi(y)$, gives

$$\begin{aligned} [z^n] \log \Phi(y(z)) &= \frac{1}{n} [x^{n-1}] L'(x) \Phi(x)^n \\ &= \frac{1}{n^2} [x^{n-1}] \frac{d}{dx} \Phi(x)^n = \frac{A_{\alpha,\beta}(n)}{n}. \end{aligned}$$

Since $y/z = \Phi(y)$, it follows that

$$\log \frac{y(z)}{z} = \sum_{n \geq 1} \frac{A_{\alpha,\beta}(n)}{n} z^n. \quad (6.2)$$

Differentiating and using (6.1) yields

$$H_{\alpha,\beta}(z) = \frac{zy'(z)}{y(z)} = \frac{1 - y^2}{1 - (\alpha - \beta)y - (\alpha + \beta + 1)y^2}. \quad (6.3)$$

For integer parameters, Φ is rational. Thus (6.1) becomes a polynomial equation after clearing denominators, and $H_{\alpha,\beta}$ is algebraic.

6.2 The target quartic

For $(\alpha, \beta) = (2, 1)$,

$$z = y(1 - y)^2(1 + y) = y - y^2 - y^3 + y^4, \quad H(z) = \frac{1 - y^2}{1 - y - 4y^2}. \quad (6.4)$$

The expanded form on the left has linear coefficient 1, so the formal inverse $y(z) \in z\mathbb{Z}[[z]]$ exists and is unique; this is the uniqueness already noted after (6.1), made explicit for the target. Eliminating y gives

$$\begin{aligned} z^2(4H - 1)^4 + zH(107H^3 - 107H^2 + 36H - 4) \\ + 32H^3(1 - H) = 0. \end{aligned} \quad (6.5)$$

Substitution of (6.4) verifies the identity directly. The partial derivative of the left side with respect to H , at $(z, H) = (0, 1)$, is -32 . Thus the formal implicit-function theorem singles out the unique branch with $H(0) = 1$. In particular, no ambiguous choice of an algebraic root is hidden in the generating-function description.

Collecting (6.5) by powers of H instead of by powers of z gives the equivalent grouped form

$$\begin{aligned} 0 = (256z^2 + 107z - 32)(H^4 - H^3) \\ + (96z^2 + 36z)H^2 - (16z^2 + 4z)H + z^2, \end{aligned} \quad (6.6)$$

which is the shape in which the elimination is easiest to check by hand. Expanding both displays shows them to be the same polynomial; the symbolic program verifies this as well.

The symbolic verification programs check the elimination in two independent ways — that the resultant of the two defining equations equals the displayed quartic, and that the quotient of the resultant by it is a nonzero constant, recorded as 1 — together with the parametrized substitution and the series identity through degree 20. The exact-arithmetic program independently verifies, in integer arithmetic alone and to degree 60, the inverse defining polynomial $y - y^2 - y^3 + y^4 = z$, the parametric form $(1 - y - 4y^2)H = 1 - y^2$, and the quartic itself. The coefficient recurrence (2.3) supplies the series in both cases, so neither check merely substitutes one rearranged symbolic expression into another.

7 A boundary test for periodic-coefficient arguments

7.1 What the logarithmic method actually requires

The odd-prime proof used two different ingredients: the equality $c_{pj} = c_j$ and the much stronger quadratic cancellation in Lemma 3.5. The first does not imply the second. Remark 3.7 states the exact sufficient hypothesis; this section shows that it is not automatic.

In detail, let $G \in 1 + xR_p[[x]]$ and suppose

$$\log G(x) = \sum_{j \geq 1} c_j x^j / j, \quad c_j \in R_p, \quad c_{pj} = c_j.$$

Define $A(n) = [x^n]G(x)^n$ and the reduced logarithm as in (3.6). The factorization and the vanishing linear term still hold. For odd p , all terms of degree at least two in the exponential have valuation at least $2v_p(N)$, so they yield a quadratic tower. A cubic tower requires the additional estimate (3.12) on the coefficients of the square of the reduced logarithm.

At the first layer and for $p \geq 5$, this gives a particularly transparent necessary and sufficient residue condition:

$$A(p) - A(1) \equiv -\frac{p^2}{2} \sum_{j=1}^{p-1} \frac{c_j c_{p-j}}{j^2} \pmod{p^3}. \quad (7.1)$$

Indeed, the quadratic coefficient at degree p is unaffected by positive powers of x^p from $\Phi(x^p)$, and $1/(j(p-j)) \equiv -1/j^2 \pmod{p}$. This proves the formula without any periodicity assumption beyond $c_{pj} = c_j$.

7.2 A spacing-three counterexample

Replace the even-occupancy restriction by occupancy divisible by 3. More generally, for a positive integer t put

$$b_t(n) = [x^n](1-x)^{-tn}(1-x^3)^{-tn}, \quad b(n) = b_1(n). \quad (7.2)$$

The logarithmic coefficients of $(1-x)^{-t}(1-x^3)^{-t}$ are $c_j = t(1 + 3\mathbf{1}_{3|j})$. They have period three and satisfy $c_{5j} = c_j$, so the exact logarithmic splitting of Section 3 still works at $p = 5$ and the linear term still vanishes. The failure is elsewhere. Nevertheless, at $t = 1$,

$$\begin{aligned} b(1) &= 1, \\ b(5) &= \binom{9}{5} + 5 \binom{6}{2} = 126 + 75 = 201, \\ b(5) - b(1) &= 200 \not\equiv 0 \pmod{125}. \end{aligned}$$

Note that $5 \nmid 3$: the failure is not caused by the prime dividing the period.

First witness: the obstruction formula. The obstruction in (7.1) is exactly

$$\sum_{j=1}^4 \frac{c_j c_{5-j}}{j^2} = 1 + \frac{4}{4} + \frac{4}{9} + \frac{1}{16} = \frac{361}{144} \equiv 4 \pmod{5}. \quad (7.3)$$

Equation (7.1) predicts the residue 75 modulo 125, which agrees with $200 \pmod{125}$.

Second witness: the quadratic coefficient itself. The same failure can be certified without passing through (7.1), by computing the quantity that Remark 3.7 actually requires. At $t = 1$ and $p = 5$,

$$[x^5] \left(\sum_{5 \nmid j} \frac{c_j}{j} x^j \right)^2 = 2 \left(1 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{4}{3} \right) = \frac{11}{6} \not\equiv 0 \pmod{5}, \quad (7.4)$$

the two products coming from the index pairs $(1, 4)$ and $(2, 3)$. Since $v_5(5) = 1$, hypothesis (3.12) demands divisibility by 5 and is violated. What this second route buys is directness: it exhibits the failure in the precise quantity the proof consumes, with no appeal to the first-level residue formula (7.1) and hence no reliance on the reduction $1/(j(p-j)) \equiv -1/j^2$. The two computations are consistent: the quantity reduced in (3.7) is the negative of the one in (7.3), and indeed $11/6$ and $-361/144$ are both congruent to 1 modulo 5.

An infinite family of first-level failures. The example is not isolated. For t with $5 \nmid t$, the factor $(1 - x^3)^{-5t}$ can contribute only its constant term or its term $5tx^3$ in degree 5, so

$$b_t(5) = \binom{5t+4}{5} + 5t \binom{5t+1}{2}, \quad b_t(1) = t.$$

Theorem 1.2 applied at $(\alpha, \beta) = (t, 0)$, where (3.14) gives $A_{t,0}(5) = \binom{5t+4}{5}$ and $A_{t,0}(1) = t$, yields $\binom{5t+4}{5} \equiv t \pmod{125}$. Therefore

$$b_t(5) - b_t(1) \equiv 5t \binom{5t+1}{2} = \frac{25t^2(5t+1)}{2} \equiv \frac{25t^2}{2} \pmod{125}, \quad (7.5)$$

the denominator 2 being read as a unit modulo 125. Hence $v_5(b_t(5) - b_t(1)) = 2$ exactly, for every t not divisible by 5. The members $t = 1, 3, 9$ give the differences 200, 13 425 and 1 953 450, with residues 75, 50 and 75 modulo 125. The member $t = 3$ is the one needed in Appendix C, because its logarithmic coefficient sequence satisfies the rational 2-function conditions at *every* prime, including 3.

Thus the success of period two is structural, not an automatic consequence of rationality, periodicity, or Frobenius invariance. In the period-two proof, the weighted square sum collapses to two ordinary reciprocal-square sums, as Remark 3.6 explains. That special collapse is absent here.

7.3 Scope of the literature correction

Müller's arXiv version 1 of *Wolstenholme Type Congruences and Framing of Rational 2-Functions* states a general periodic weighted-harmonic bound in Theorems 1.2 and 6.2, a corresponding precise cubic framing bound in Theorem 1.1, and the framing coefficient formula in Proposition 5.2(3) [7]. The value (7.3) contradicts the printed harmonic bound at period 3 and prime 5.

The coefficient counterexample also fits the printed number-field hypotheses, not merely a superficial formal analogy. Take $K = \mathbb{Q}(\sqrt{-3})$, whose discriminant is -3 , and

$$V_1(x) = \frac{x}{1-x} + \frac{3x^3}{1-x^3}, \quad c_j = 1 + 3\mathbf{1}_{3|j}. \quad (7.6)$$

At every unramified rational prime $p \neq 3$, its rational coefficients satisfy $c_{mp^r} = c_{mp^{r-1}}$ exactly. Thus it is a rational 2-function in the convention of that preprint. Proposition 5.2(3) there identifies the coefficient of its plus-framing with parameter one as $[x^n] \exp(n \int V_1) = b(n)$, where the formal integration sends x^j to x^j/j . Prime 5 is unramified and does not divide the period. The precise bound in Theorem 1.1 would therefore give divisibility by 125, contrary to $b(5) - b(1) = 200$.

Appendix C carries out the same audit with a second, independent input V_3 defined over $\mathbb{Q}$ alone, for which the rational 2-function condition is proved at *every* prime, with no ramified exception and no field extension. Both witnesses are retained: the one here is smaller, the one there has no hypothesis that has to be read off a discriminant. The subscripts distinguish them throughout; note in particular that their weighted harmonic sums, $361/144$ and $361/16$, are different numbers computed for different inputs, and neither is a misprint for the other.

A different period-four counterexample is already documented by the same repository that supplied the prior odd-prime proof [4]. Accordingly, the discovery of a problem in the general framing statement is not claimed as new here. The spacing-three examples are compact independent boundary witnesses. The criticism is specifically of the printed uniform bounds

in version 1. A single exceptional prime does not refute a weaker assertion that permits an unspecified finite set of additional exceptional primes, or one with a compensating scalar, and it says nothing by itself about unrelated framing results. None of the proofs in this article uses the disputed statements.

8 Asymptotic expansion of the basket sequence

The leading asymptotic is recorded in the OEIS entry, attributed to Kotesovec [1]. Here we derive it together with two explicit correction terms and an all-order coefficient formula. The method is the standard saddle-point approach to large powers of generating functions [8, Chapter VIII].

8.1 The saddle and exponential growth

For this section only, put

$$\Phi(x) = \frac{1}{(1-x)(1-x^2)}.$$

The positive saddle $\tau \in (0, 1)$ satisfies

$$\tau \frac{\Phi'(\tau)}{\Phi(\tau)} = 1.$$

Since $x\Phi'/\Phi = x/(1-x) + 2x^2/(1-x^2)$, this is equivalent to $4\tau^2 + \tau - 1 = 0$. Hence

$$\tau = \frac{\sqrt{17}-1}{8}, \quad \rho = \frac{\tau}{\Phi(\tau)} = \frac{51\sqrt{17}-107}{512}, \quad \lambda = \rho^{-1} = \frac{107+51\sqrt{17}}{64}. \quad (8.1)$$

Numerically, $\lambda = 4.9574747954140732506\dots$. Define the logarithmic cumulants

$$\kappa_j = \left(x \frac{d}{dx} \right)^j \log \Phi(x) \Big|_{x=\tau}. \quad (8.2)$$

Then $\kappa_1 = 1$ and

$$\begin{aligned} \kappa_2 &= \frac{51 - 5\sqrt{17}}{16}, & \kappa_3 &= \frac{435 - 69\sqrt{17}}{32}, \\ \kappa_4 &= \frac{10731 - 2125\sqrt{17}}{128}, & \kappa_5 &= \frac{91545 - 20175\sqrt{17}}{128}, \\ \kappa_6 &= \frac{4008567 - 929185\sqrt{17}}{512}. \end{aligned} \quad (8.3)$$

Theorem 8.1 (Two correction terms). *As $n \rightarrow \infty$ through positive integers,*

$$a(n) = \frac{\lambda^n}{\sqrt{2\pi\kappa_2 n}} \left(1 + \frac{c_1}{n} + \frac{c_2}{n^2} + O(n^{-3}) \right), \quad (8.4)$$

where

$$c_1 = \frac{-1683 + 95\sqrt{17}}{9248}, \quad c_2 = \frac{119821 - 28605\sqrt{17}}{5030912}. \quad (8.5)$$

The leading amplitude is equivalently

$$\frac{1}{\sqrt{2\pi\kappa_2}} = \frac{\sqrt{3+5/\sqrt{17}}}{4\sqrt{\pi}}. \quad (8.6)$$

8.2 Derivation, including control of the remainder

Cauchy's coefficient formula on the circle $|x| = \tau$ gives

$$a(n) = \frac{\lambda^n}{2\pi} \int_{-\pi}^{\pi} \exp \left(n \left\{ \log \frac{\Phi(\tau e^{i\theta})}{\Phi(\tau)} - i\theta \right\} \right) d\theta. \quad (8.7)$$

The coefficients of Φ are positive and its support contains 0 and 1. Equality in the triangle inequality $|\Phi(\tau e^{i\theta})| \leq \Phi(\tau)$ therefore requires $e^{i\theta} = 1$. On every closed subinterval bounded away from zero, the integrand is exponentially smaller than the central contribution.

Near zero, choose the analytic logarithm agreeing with the real logarithm at τ . The saddle condition cancels the linear term, and the exponent inside braces is

$$-\frac{\kappa_2 \theta^2}{2} + \sum_{k \geq 3} \frac{\kappa_k (i\theta)^k}{k!}. \quad (8.8)$$

In particular, its real part is at most $-c\theta^2$ for sufficiently small $|\theta|$ and some $c > 0$.

Set $u = \sqrt{n\kappa_2} \theta$. The central integral becomes a Gaussian integral multiplied by

$$\exp \left(\sum_{k \geq 3} \frac{\kappa_k i^k u^k}{k! \kappa_2^{k/2}} n^{1-k/2} \right).$$

Expanding this expression and integrating term by term is justified at any fixed order by analytic Taylor remainder estimates and Gaussian domination. For specificity, split a small fixed central interval at $|\theta| = n^{-2/5}$. Outside that shrinking interval the bound $e^{-cn\theta^2}$ is exponentially small in $n^{1/5}$. Inside it, Taylor expansion to sufficiently high finite degree has a uniform remainder; rescaling and integrating its polynomial bounds against a slightly wider Gaussian gives the stated power remainder. Odd powers have zero symmetric Gaussian moment. Carrying the expansion two orders further than the last retained even power bounds the relative remainder by $O(n^{-3})$ in (8.4). The same argument works to every fixed order.

Using

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} u^{2\ell} e^{-u^2/2} du = (2\ell - 1)!!,$$

one obtains

$$\begin{aligned} c_1 &= \frac{\kappa_4}{8\kappa_2^2} - \frac{5\kappa_3^2}{24\kappa_2^3}, \\ c_2 &= -\frac{\kappa_6}{48\kappa_2^3} + \frac{7\kappa_3\kappa_5}{48\kappa_2^4} + \frac{35\kappa_4^2}{384\kappa_2^4} \\ &\quad - \frac{35\kappa_3^2\kappa_4}{64\kappa_2^5} + \frac{385\kappa_3^4}{1152\kappa_2^6}. \end{aligned} \quad (8.9)$$

Substitution of (8.3) simplifies to (8.5); elementary rationalization gives (8.6). This proves the theorem.

8.3 An explicit all-order coefficient formula

The same calculation gives a finite expression for every correction coefficient. For $j \geq 0$, sum over nonnegative integers $m_3, \dots, m_{2j+2}$ satisfying

$$\sum_{k=3}^{2j+2} (k-2)m_k = 2j, \quad K = \sum_{k=3}^{2j+2} km_k.$$

The integer K is even. Then

$$c_j = \sum (-1)^{K/2} (K-1)!! \kappa_2^{-K/2} \prod_{k=3}^{2j+2} \frac{\kappa_k^{m_k}}{m_k! (k!)^{m_k}}. \quad (8.10)$$

For $j = 0$, the empty product and $(-1)!!$ are both 1, giving $c_0 = 1$. The constraint comes from collecting the exponent of $n^{-1/2}$ in the expanded exponential; the double factorial comes from its Gaussian moment. This is a direct finite enumeration formula, not a recurrence with unknown initial correction terms.

Numerically, $c_1 \approx -0.1396307272454933$ and $c_2 \approx 0.0003736029529450764$. The cancellation in c_2 is substantial, so exact symbolic simplification is preferable to low-precision arithmetic. The supplied numerical checks use 90 decimal digits. For illustration, the signed relative error of the approximation through c_2 is about $-9.6264 \cdot 10^{-9}$ at $n = 100$ and $-6.1254 \cdot 10^{-10}$ at $n = 250$. Those numerical comparisons are checks, not the proof of the remainder.

9 Reproducible computation and proof audit

9.1 What the programs compute

The archive separates exact integer checks from optional symbolic and high-precision checks. The program `verify.py` uses only the Python standard library and implements (2.3); it cross-checks the target against (2.1), against (2.2), and against a fourth, wholly independent mechanism, namely repeated convolution of the single-pair basket series $\lfloor j/2 \rfloor + 1$. It also tests Lemmas 3.1, 3.2 and 3.5 directly, the boundary family (7.2), the rational 2-function hypothesis used in Appendix C, the Möbius denominators, and the generating-function identities of Section 6 in integer arithmetic to degree 60. The program `binary_checks.py` tests the complete leading-defect formula, the harmonic-unit lemma, and both common-denominator assertions. The program `symbolic_checks.py` uses SymPy and mpmath for elimination, Gaussian coefficient enumeration, and asymptotic comparisons.

The recorded run produced the following exact checks. Each count is a finite coverage statement, not an extrapolation to all indices.

Check	Cases
Initial target terms, against the inspected OEIS listing	26
Target recurrence versus positive binomial sum	208
Target recurrence versus independent basket convolution	13
Signed parameter recurrence versus binomial sum	1,519
Target odd-prime congruences, strong form	489
Two-parameter odd-prime congruences, strong form	3,675
Odd-prime quadratic-tail tests	918
Logarithmic-derivative coefficient bounds	3,415
Factorial inequalities for the exponential tail	119,760
Generating-function coefficients, integer arithmetic	61
Original Möbius/orbit checks	300
Cubic Möbius denominator indices	100
Exact boundary witnesses	7
Period-three boundary family members	3
Rational 2-function input checks for the audit	10,000
Complete binary leading-defect formula	5,808
Binary first-lift congruence	1,936
Exact binary harmonic-unit normalization	400
Target denominator-12 and orbit divisibility	1,000
General denominator-24 / conditional denominator-12	4,900
Binary sharpness and minimal-denominator witnesses	290

All these checks passed, in a single run of the merged programs; no count above is spliced from a different sweep. The binary leading-defect grid covers $-5 \leq \alpha, \beta \leq 5$, odd $1 \leq m \leq 15$, and $2 \leq r \leq 7$, with $m2^r \leq 2000$. The target odd-prime run uses odd primes at most 97, multipliers at most 12 *including* those divisible by the prime, and indices at most 5000; the largest index actually reached is 5000 itself, at $p = 5$, $m = 8$, $r = 4$. Because multipliers divisible by p are no longer skipped, the run exercises the strong exponent $3(r + v_p(m)) - \varepsilon_p$ of (1.8), not merely the coprime case. The general odd-prime grid uses parameters from -3 to 3 , primes 3, 5, 7, 11, 13, and indices at most 350. Zero differences are recorded as having infinite valuation. The derivative and quadratic checks deliberately include signed and degenerate parameter pairs, for which some prescribed bounds are zero and some coefficients vanish; those cases are counted rather than silently discarded.

The symbolic run additionally verifies the logarithmic derivative, the quartic resultant together with the constant quotient of that resultant by the displayed quartic, its parametrization, its power series through degree 20, the three parity-weight polynomial identities used in Lemma 3.5, the saddle point, the growth constant, the leading amplitude, and the Gaussian formulas for c_1 and c_2 . These tests used Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0 in the recorded environment; the exact scripts are not tied to those library versions.

9.2 Reproduction commands and output data

From the archive directory, run

```
python verify.py
python binary_checks.py
python -m pip install -r requirements-optional.txt
python symbolic_checks.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Only the third and fourth commands need optional Python packages. No network access is used

by any of the checking programs themselves, and no floating-point arithmetic occurs in any congruence check: every such test is performed in exact integer, rational, or modular arithmetic. The integer checker accepts command-line coverage parameters; `-max-n 20000` extends the original-sequence sweep, and `python verify.py -help` lists the rest. In PowerShell the script path may also be written `python .\verify.py`.

The CSV files contain target terms through 1000, the exact prime-power valuation checks, the full binary parameter grid, primitive-orbit and Möbius values, the period-three boundary family, and asymptotic comparisons. The reports preserve the actual check counts. There are no randomized tests and no unrecorded dependence on an external computer-algebra session.

9.3 Source and artifact notes

Sources were inspected on 19 September 2026. The framing theorem statements of [7] were checked against the PDF of arXiv:2104.10754v1 — page 3 for Theorem 1.1 and page 4 for Theorem 1.2, with Proposition 5.2(3) for the framing coefficient formula — and not only against the HTML rendering, because the two can differ in the precise quantifiers. The prior proof candidate and the prior period-four counterexample [3, 4] were read at a pinned immutable commit, whose full hash appears in the bibliography. No database edit, no OEIS submission, and no communication with any author was performed. No source PDFs, third-party repository trees, or font files are bundled with this archive.

9.4 Independent points of possible failure

Several interfaces deserve explicit scrutiny in any independent review. First, a recurrence in the coefficient index must not be mistaken for a recurrence in the diagonal index. Second, the odd-prime exponential has a vanishing linear term, but the binary one does not; the latter is handled by reflection, not by reusing the former proof. Third, the $1/2$ in formal integration by parts at the binary prime is retained throughout and is responsible for one of the losses. Fourth, the Möbius grouping is performed at the exact prime-power exponent of n , not merely at a prime dividing n . Finally, the finer binary valuation table is not used to justify any of the uniform bounds.

The article uses only elementary formal series, local rational arithmetic, coefficient extraction, and the stated saddle-point analysis. It does not invoke the disputed periodic framing bound or any unverified asymptotic recurrence from an OEIS comment.

10 Further questions after the binary completion

The uniform small-prime completion is proved, but it does not determine all exceptional extra divisibility. For the basket sequence, the observed valuations at $n = 2^r$ are

r	1	2	3	4	5	6	7	8
$v_2(a(2^r) - a(2^{r-1}))$	2	4	8	13	16	19	22	25

The recorded exact computation continues this pattern through $r = 15$.

Conjecture 10.1 (A finer target valuation, not used above). For every $r \geq 4$,

$$v_2(a(2^r) - a(2^{r-1})) = 3r + 1.$$

This is not a consequence of Theorem 1.4, whose leading residue vanishes for $(\alpha, \beta) = (2, 1)$. Proving it would require following the reflection calculation several binary digits further, including the next residues of $K(t)$ and additional exponential terms. The statement for a fixed odd multiplier is more varied: at $m = 5$ the values from $r = 4$ onward in the checked range are $3r + 3$, while at $m = 7$ they are $3r + 2$.

A second question is to characterize the periodic logarithmic coefficient sequences for which the cubic tail estimate holds. Formula (7.1) gives an immediate first-level obstruction, but vanishing at the first level alone does not establish all higher levels. The spacing-three example shows why a condition stronger than periodicity and Frobenius invariance is needed: any classification of the coefficient patterns satisfying the criterion (3.12) must accommodate the period-three obstruction rather than assume it away. A useful classification would separate identities forced by symmetry from congruences that depend on the prime.

Closely related, and equally open, is the question of strictness. Neither Theorem 1.2 nor Theorem 1.4 classifies the parameters, primes and levels at which its lower bound is attained rather than exceeded. Table 1 shows equality throughout its odd range, and Theorem 1.4 settles the binary case for odd $\alpha\beta$ and $m = 1$; between these the picture is empirical.

A third question is to refine the common denominator for a fixed parameter pair. The bounds 24 and 12 are optimal uniformly over the stated classes, but special pairs can have smaller denominators. The exact first binary residue already detects one such improvement. An analogous explicit ternary residue could identify further parameter-dependent reductions.

These questions are distinct from the completed claims. In particular, the remaining uncertainty about the finer valuation pattern does not limit the proof of the universal denominator-12 theorem.

11 Conclusion

Two theorems have been proved, by two different mechanisms. At an odd prime, the parity of the logarithmic coefficients survives multiplication by p , which produces an exact factorization; the whole difficulty is then concentrated in one weighted reciprocal-square sum, and Lemmas 3.3 and 3.5 show that it supplies the third power of p , with a single unavoidable loss at $p = 3$ traceable to the valuation of the constant 3. At $p = 2$ that parity argument is unavailable, and the binary completion instead converts a linear exponential contribution into a quadratic one by reflection and then extracts an extra index factor by formal integration by parts. An exact odd-harmonic normalization identifies the first possible binary bit, not just a lower bound. Priority for the odd-prime statement belongs to the prior proof candidate located in the source audit; the binary results were not found in the checked sources.

For A348410 the completed all-prime estimate is naturally encoded by the single integer 12. It clears every denominator of the cubic Möbius transform, is forced by the values at $n = 3$ and $n = 4$, and is already attained in full at the single index $n = 12$. The resulting primitive cyclic-word divisibility gives a combinatorial expression of the same arithmetic fact.

The negative results are worth keeping in view beside the positive ones. Exact invariance of the logarithmic coefficients under index multiplication is not enough; the quadratic estimate (3.12) is a separate hypothesis, and an infinite period-three family violates it at $p = 5$. The same family refutes the explicit conclusion of a printed general framing statement, which is why no proof here cites that statement.

The archive contains the full proofs and reproducible checks; priority beyond the audited sources and external validation of the new binary argument remain matters for further review.

A A minimal exact evaluator

The following code is the computational core; the full scripts add input validation, independent formula checks, report generation, and CSV output.

```
def coefficient_diagonal(n: int, alpha: int = 2,
                        beta: int = 1) -> int:
    if n < 0:
        raise ValueError("n must be nonnegative")
    f_minus_two, f_minus_one = 0, 1
    for k in range(1, n + 1):
        numerator = n * (alpha - beta) * f_minus_one
        numerator += (n * (alpha + beta) + k - 2) * f_minus_two
        current, remainder = divmod(numerator, k)
        if remainder:
            raise ArithmeticError("coefficient division was not exact")
        f_minus_two, f_minus_one = f_minus_one, current
    return f_minus_one
```

The running names refer to the two previously available coefficients at the start of each iteration. At $n = 0$ the function correctly returns 1.

B Notation and dependency map

Symbol	Meaning
$A_{\alpha,\beta}(n)$	Integer-parameter coefficient diagonal in (1.4).
$a(n)$	The specialization $A_{2,1}(n)$, OEIS A348410.
$A^{(d)}(n)$	The change-of-step family of Corollary 3.8.
ε_p	The ternary loss (1.6): 1 at $p = 3$, else 0.
R_p	Rationals with denominator prime to p .
c_j	Logarithmic-derivative coefficients, (3.1).
g_j	Coefficients of a power G^M , Lemma 3.1.
D_p	Reduced odd-prime logarithm in (3.6).
$H_p(s)$	Reduced reciprocal-square sum; not the generating function.
$T_p(s)$	Quadratic tail coefficient $[x^s]D_p^2$, (3.7).
$h(T)$	Sum of reciprocals of the positive odd integers less than T .
U, K, V	Binary series in (4.1) and (4.4); V is local to Section 4.
$b_t(n)$	Period-three boundary family (7.2); $b = b_1$.
V_1, V_3	The two framing audit inputs, (7.6) and (C.3).
$B(n)$	Cubic Möbius transform $n^{-3} \sum_{d n} \mu(d)a(n/d)$.
P_n	Primitive cyclic-word count, equal to $n^2 B(n)$.
$H(z)$	Ordinary generating function of $a(n)$.
κ_j	Saddle cumulants in (8.2).

The proof dependencies are short. Lemmas 3.1–3.5 imply Theorem 1.2, and hence Corollary 1.3. Lemma 4.1, reflection, and (4.12) imply Theorem 1.4. These two theorems plus the first binary lift imply Theorem 1.5. Grouping Möbius divisors then proves Theorem 1.6; periodic word decomposition gives Corollary 1.7. The generating-function and asymptotic sections are independent checks and complementary structure, not premises of the arithmetic proofs. Appendix C is a negative result about an external statement and is used by nothing above it.

C Audit of a broader framing claim

An apparently relevant external source is Müller's preprint [7], arXiv:2104.10754v1. Its general statements, if correct as printed, would imply a cubic prime-power congruence for a much wider class than Theorem 1.2 covers, and it would be tempting to cite them as a black box in place of Section 3. This appendix records why we do not. Nothing in the article depends on it.

In the coefficient convention of that preprint, a *rational 2-function* over $\mathbb{Q}$ is a rational series $V(x) = \sum_{j \geq 1} c_j x^j$ with integral coefficients satisfying

$$c_{mp^r} \equiv c_{mp^{r-1}} \pmod{p^{2r}} \quad \text{for every prime } p \text{ and all } m, r \geq 1. \quad (\text{C.1})$$

For framing parameter 1, Proposition 5.2(3) there gives the transformed coefficients

$$\text{fr}(n) = [x^n] \exp\left(n \sum_{j \geq 1} \frac{c_j}{j} x^j\right), \quad (\text{C.2})$$

and the explicit $p \geq 5$ conclusion of its Theorem 1.1 asserts cubic prime-power congruences for fr whenever p does not divide the coefficient period.

Section 7.3 already exhibited one input satisfying the printed hypotheses and violating that conclusion, namely V_1 of (7.6), read over $K = \mathbb{Q}(\sqrt{-3})$. The witness below is independent of it and is kept because it buys something the first does not: its rational 2-function property is proved at every prime by a finite case split, with no ramified-prime exception and no field extension anywhere in the argument.

C.1 An input that satisfies the condition at every prime

Take

$$V_3(x) = \frac{3x}{1-x} + \frac{9x^3}{1-x^3}, \quad c_j = 3 + 9 \cdot \mathbf{1}_{3|j}. \quad (\text{C.3})$$

The coefficient sequence is integral and has period 3. We verify (C.1) for all indices, not merely on a finite test range. Let p be a prime and $m, r \geq 1$.

- If $p \neq 3$, then multiplication by p preserves divisibility by 3, so $3 \mid mp^r$ if and only if $3 \mid mp^{r-1}$, and the two coefficients are *equal*.
- If $p = 3$ and $r \geq 2$, both $m3^r$ and $m3^{r-1}$ are divisible by 3, so again the two coefficients are equal.
- If $p = 3$ and $r = 1$, the difference $c_{3m} - c_m$ is 0 when $3 \mid m$ and 9 otherwise. In both cases it is divisible by $3^2 = p^{2r}$.

Hence V_3 is a rational 2-function over $\mathbb{Q}$ in the stated convention, with no ramified-prime exception and no passage to a number field.

C.2 The refutation arithmetic

Formal integration of (C.3) gives

$$\sum_{j \geq 1} \frac{c_j}{j} x^j = -3 \log(1-x) - 3 \log(1-x^3),$$

so that (C.2) becomes, in the notation of (7.2),

$$\text{fr}(n) = [x^n] (1-x)^{-3n} (1-x^3)^{-3n} = b_3(n).$$

At $n = 1$ and $n = 5$,

$$\begin{aligned}\mathrm{fr}(1) &= 3, \\ \mathrm{fr}(5) &= \binom{19}{5} + 15 \binom{16}{2} = 11\,628 + 1\,800 = 13\,428, \\ \mathrm{fr}(5) - \mathrm{fr}(1) &= 13\,425 = 25 \cdot 537 \equiv 50 \pmod{125}.\end{aligned}$$

The prime 5 is unramified over $\mathbb{Q}$ and does not divide the period 3, so Theorem 1.1 as printed would require divisibility by $5^3 = 125$. It fails. This is the member $t = 3$ of the family (7.5), whose exact valuation there is 2.

C.3 The weighted harmonic assertion fails on the same input

Theorem 1.2 of the same preprint, repeated there as Theorem 6.2, would require the period-three weighted sum below to vanish modulo 5 at $p = n = 5$. For the very same coefficients (C.3), exact rational arithmetic gives

$$\sum_{k=1}^4 \frac{c_{5-k} c_k}{k^2} = 9 + \frac{36}{4} + \frac{36}{9} + \frac{9}{16} = \frac{361}{16} \equiv 1 \pmod{5}, \quad (\text{C.4})$$

and the corresponding quadratic coefficient is $33/2$, which likewise does not vanish modulo 5.

The value $361/16$ here and the value $361/144$ of (7.3) are computed for *different* inputs — V_3 with $c_j = 3 + 9 \cdot \mathbf{1}_{3|j}$, and V_1 with $c_j = 1 + 3 \cdot \mathbf{1}_{3|j}$ respectively — and their residues modulo 5 differ accordingly, 1 against 4. Neither is a misprint for the other. Both are checked exactly by the supplied programs.

C.4 Scope of this appendix

The scope of the observation matters, and it is narrow. A failure at one prime refutes the cited explicit all-eligible-primes conclusion, as printed in the inspected version. It does not by itself refute a weaker statement that allows an unspecified finite set of exceptional primes, or a compensating scalar, and it says nothing about the author’s intentions, about an uninspected later revision, or about unrelated framing results. A period-four counterexample is in any case already documented elsewhere [4], so no novelty is claimed for the observation itself.

This audit is included because, without it, it would be tempting to cite the framing statement as a black box. The main proof does not do that. Its logarithmic expansion and coefficient estimate are in the same mathematical tradition, but the reciprocal-square statement it needs is proved directly for parity weights in Lemmas 3.3 and 3.5. That replacement is sufficient for A348410 and its integer-parameter family, and it does not inherit the false arbitrary-period assertion.

References

- [1] OEIS Foundation Inc., *A348410: Number of nonnegative integer solutions to $n = \sum_{i=1}^n (a_i + b_i)$, with b_i even*. Includes P. Bala’s 21 February 2022 coefficient formulas and supercongruence conjecture, and V. Kotesovec’s leading asymptotic. <https://oeis.org/A348410>. Inspected 19 September 2026.
- [2] OEIS Foundation Inc., *A352373*. Related parameterized coefficient-power congruence. <https://oeis.org/A352373>. Inspected 19 September 2026.

- [3] GitHub repository `rbajaj5/a183068-supercongruence`, *A cubic tower for a two-parameter coefficient family*, `related-results/CoefficientFramingCubicTower.md`. Public proof candidate; pinned repository snapshot 3085b46fdbce (full hash in the linked URL). [Permanent link to the inspected note](#). Inspected 19 September 2026. The source states no binary assertion.
- [4] GitHub repository `rbajaj5/a183068-supercongruence`, *Counterexample to the rational-framing theorem*, `related-results/RationalFramingCounterexample.md`, same pinned snapshot as [3]. [Permanent link to the period-four counterexample](#). Inspected 19 September 2026; also cross-checked against the source’s result index and coefficient-framing discussion.
- [5] T. Niu, *An explicit algebraic generating function for OEIS A348410*, arXiv:2605.16553, 2026. <https://arxiv.org/abs/2605.16553>.
- [6] H. Prodinger, *The generating function of A348410 in OEIS using the diagonal method and another sequence (A001008) from OEIS*, arXiv:2605.21255, version 2, 23 May 2026. <https://arxiv.org/abs/2605.21255>.
- [7] L. Felipe Müller, *Wolstenholme Type Congruences and Framing of Rational 2-Functions*, arXiv:2104.10754, version 1, 21 April 2021. Cited loci: Theorem 1.1, Theorem 1.2 (repeated as Theorem 6.2), and Proposition 5.2(3), the framing coefficient formula. <https://arxiv.org/abs/2104.10754>. Statements checked against the PDF, pages 3 and 4, and not only the HTML rendering; inspected 19 September 2026. The criticism in Section 7 and Appendix C concerns the precise printed bounds of Theorems 1.1, 1.2, and 6.2 in this version, not all framing theory.
- [8] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, especially Chapter VIII. Authors’ book site: <https://ac.cs.princeton.edu/home/>.